Answers

1) b	**2)** b	**3)** d	**4)** a	**5)** b
6) c	**7)** d	**8)** b	**9)** c	**10)** d

Short Answer Questions

1. What is angle of deviation due to reflection?
2. Derive the relation between f and R for a spherical mirror.
3. What are the Cartesian sign conventions for spherical mirrors?
4. What is optical path? Obtain the equation for optical path.
5. State Snell's law/law of refraction.
6. What is angle of deviation due to refraction?
7. What is principle of reversibility?
8. What is relative refractive index?
9. Obtain the equation for apparent depth.
10. Why do stars twinkle?
11. What are critical angle and total internal reflection?
12. Obtain the equation for critical angle.
13. Explain the reason for the glittering of diamond.
14. What are mirage and looming?
15. Write a short note on the prisms making use of total internal reflections.
16. What is Snell's window?
17. How does an endoscope work?
18. What are primary focus and secondary focus of a lens?
19. What are the sign conventions followed for lenses?
20. Arrive at lens equation from lens maker's formula.
21. Obtain the equation for lateral magnification of thin lens.
22. What is power of a lens?
23. Derive the equation for effective focal length for lenses in contact.
24. What is angle of minimum deviation?
25. What is dispersion?
26. How are rainbows formed?
27. What is Rayleigh's scattering?
28. Why does sky appear blue?
29. What is the reason for reddish appearance of sky during sunset and sunrise?
30. Why do clouds appear white?

Long Answer Questions

1. Derive the mirror equation and the equation for lateral magnification.
2. Describe the Fizeau's method to determine the speed of light.
3. Obtain the equation for radius of illumination (or) Snell's window.
4. Derive the equation for acceptance angle and numerical aperture of optical fibre.
5. Obtain the equation for lateral displacement of light passing through a glass slab.
6. Derive the equation for refraction at single spherical surface.
7. Obtain lens maker's formula and mention its significance.
8. Derive the equations for thin lens and for magnification.
9. Derive the equation for angle of deviation produced by a prism and thus obtain the equation for refractive index of material of the prism.

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